Section 6.1

Problem 8. *Can a graph have an odd number of vertices of odd degree? Explain.*

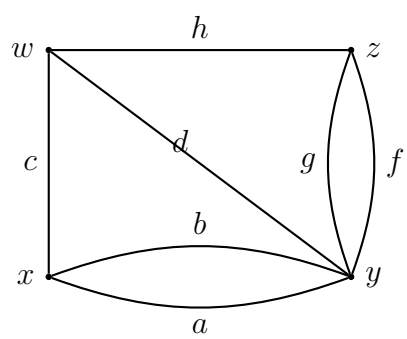
Solution. Not possible, and this is explained via the handshake lemma. If we sum up the degrees of all vertices then the resulting number is 2 multiplied by the number of edges. The number 2 multiplied by anything is by definition even. □

Problem 10. *Draw a picture of the graph G with $V(G) = \{x, y, z, w\}$, $E(G) = \{a, b, c, d, f, g, h\}$, and γ given by the following table.*

e	a	b	c	d	f	g	h
$\gamma(e)$	$\{x, y\}$	$\{x, y\}$	$\{w, x\}$	$\{w, y\}$	$\{y, z\}$	$\{y, z\}$	$\{w, z\}$

Solution.

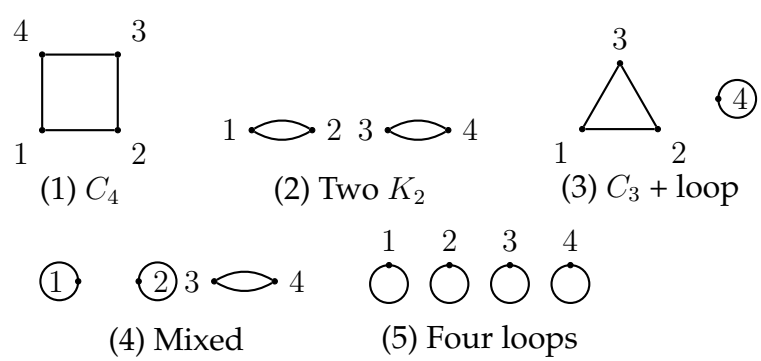
The graph has 4 vertices and 7 edges. Note that edges a and b are parallel edges between x and y, and edges f and g are parallel edges between y and z.



□

Problem 13. *(a) Draw pictures of all five of the regular graphs that have four vertices, each vertex of degree 2. "All" here means that every regular graph with four vertices and each vertex of degree 2 is isomorphic to one of the five, and no two of the five are isomorphic to each other.*

Solution.

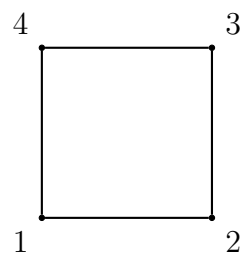


□

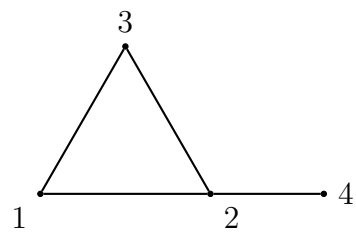
Problem 14. *(b) Draw pictures of the two graphs with four vertices and four edges that have no loops or parallel edges.*

Solution.

With 4 vertices and 4 edges (no loops or parallel edges), there are exactly two non-isomorphic graphs:



Graph 1: C_4 (4-cycle)
Degree sequence: $(2, 2, 2, 2)$



Graph 2: C_3 with tail
Degree sequence: $(2, 2, 3, 1)$

Verification:

- Graph 1 has 4 edges forming a cycle, all vertices have degree 2
- Graph 2 has 3 edges forming a triangle plus 1 edge to a 4th vertex, giving degrees $(1, 2, 2, 3)$
- These are the only two non-isomorphic simple graphs with 4 vertices and 4 edges

□

Problem 15. *Which, if any, of the pairs of graphs shown in Figure 8 are isomorphic? Justify your answer by describing an isomorphism or explaining why one does not exist.*

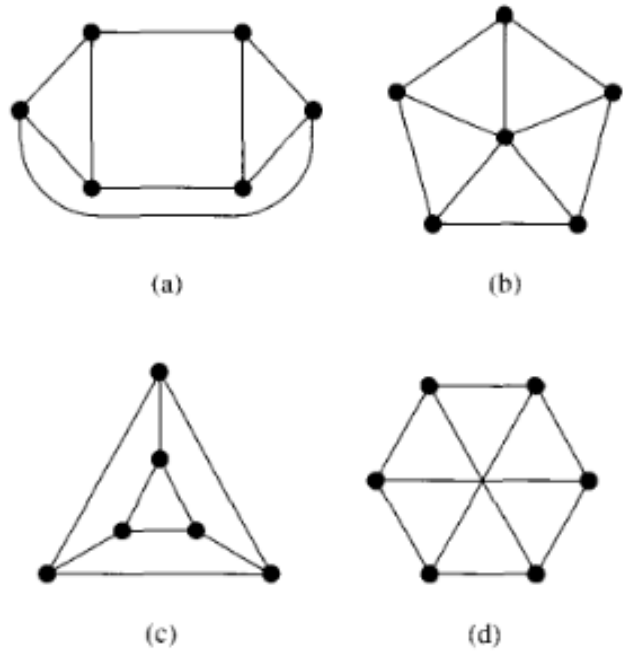


Figure 8 ▲

solution. Doesn't appear any of them are isomorphic, (a) has 2 cycles of 3 edges, and 2 cycles of 4 edges, (b) has 5 cycles of 3 edges and 1 cycle of 5 edges, (c) has 3 cycles of 4 edges and 2 cycles of 3 edges, and d 4 cycles of 4 edges and 1 of 6 edges. □

Problem 16. *Describe an isomorphism between the graphs shown in Figure 9.*

Figure 9:

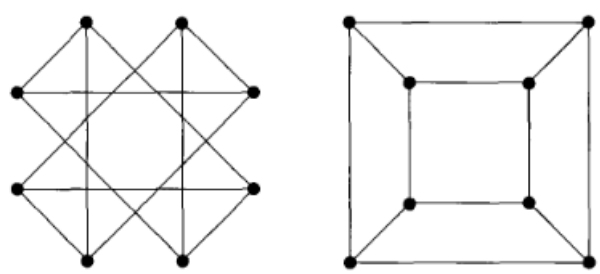


Figure 9 ▲

solution. Each vertex has 3 edges in both 8 total vertices in both, There are 12 total edges in both, 6 total cycles all of equal length(4 edges) in both, These are isomorphic, you can create a bijective mapping between these. □

Problem 20. *(a) A graph with 21 edges has seven vertices of degree 1, three of degree 2, seven of degree 3 and the rest of degree 4. How many vertices does it have?*

(b) How would your answer to part (a) change if the graph also had six vertices of degree 0?

Solution.

(a) Let k be the number of vertices of degree 4. By the Handshake Lemma, the sum of all degrees equals twice the number of edges:

$$\begin{aligned} 7(1) + 3(2) + 7(3) + k(4) &= 2(21) \\ 7 + 6 + 21 + 4k &= 42 \\ 34 + 4k &= 42 \\ 4k &= 8 \\ k &= 2 \end{aligned}$$

Total vertices: $7 + 3 + 7 + 2 = 19$ vertices.

(b) Vertices of degree 0 contribute nothing to the degree sum, so the calculation in part (a) remains unchanged. We still have 2 vertices of degree 4.

Total vertices: $7 + 3 + 7 + 2 + 6 = 25$ vertices.

□

Section 6.2

Problem 2. *(a) Give the vertex sequence of a simple closed path of largest possible length in the graph of Figure 7(a).*

(b) Is your path in part (a) an Euler circuit for the graph?

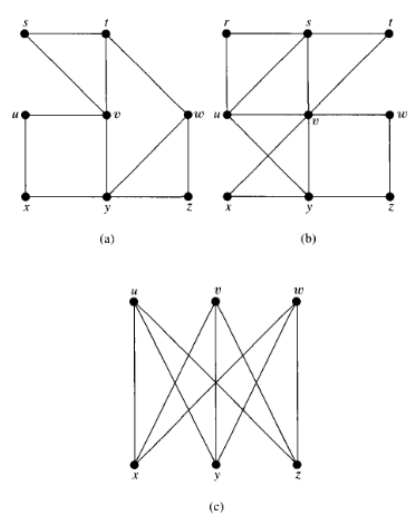


Figure 7 ▲

Solution. (a) [wzywtvyxuvstw]
(b) Yes, this is indeed a Euler Circuit □

Problem 7. *Consider the graph shown in Figure 8(a).*

(a) Describe an Euler path for this graph or explain why there isn't one.

(b) Describe an Euler circuit for this graph or explain why there isn't one.

Figure 8(a):

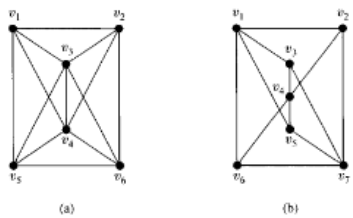


Figure 8 ▲

solution. (a) Eulerian path exists, just having trouble finding it.
(b) There is no Eulerian circuit since all vertices must have even degrees for this criteria and we have 2 vertices with odd degrees. □

Problem 8. *Repeat Exercise 7 for the graph of Figure 8(b).*

(a) Describe an Euler path for this graph or explain why there isn't one.

(b) Describe an Euler circuit for this graph or explain why there isn't one.

solution. (a) Eulerian path does not exist, there are more than two vertices with odd degree.
(b) The lack of Eulerian path would disqualify it from having an Eulerian Circuit. □

Problem 15. *An old puzzle presents a house with 5 rooms and 16 doors, as shown in Figure 9. The problem is to figure out how to walk around and through the house so as to go through each door exactly once.*

(a) Is such a walk possible? Explain.

(b) How does your answer change if the door adjoining the two large rooms is sealed shut?

Figure 9:

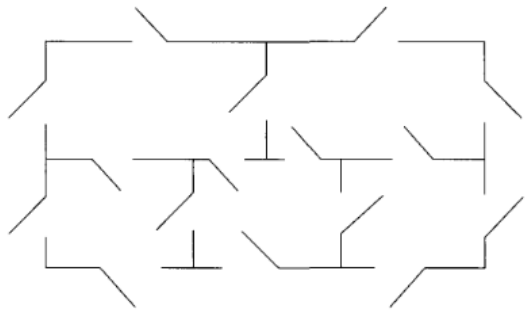


Figure 9 ▲

solution. (a) Impossible, same problems as the 6 bridges of konigsberg, there needs to be either 0 or two nodes to create a Eulerian path.
(b)It changes the answer since now only 2 odd vertices would remain if we modlled this into a graph, meaning it would be possible. □

Section 6.3

Problem 2. *The trees in Figure 6 have seven vertices. Specify which tree in Figure 3 each is isomorphic to.*

Figure 3:

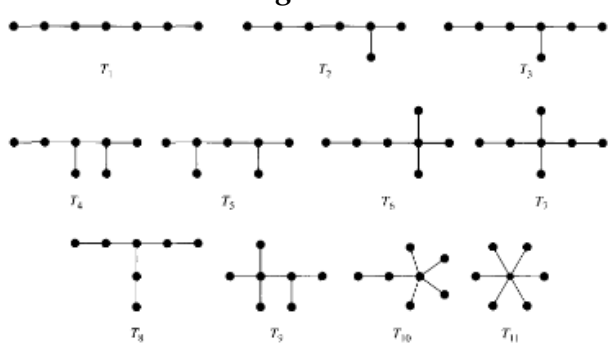


Figure 6:

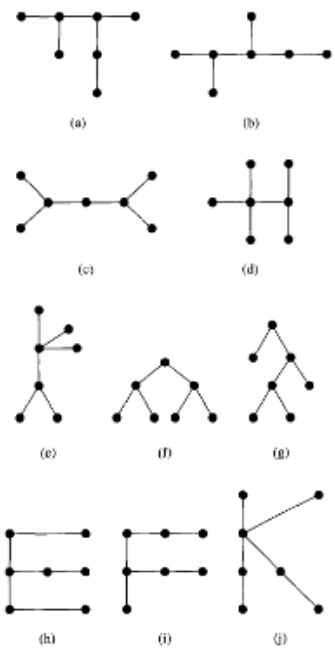
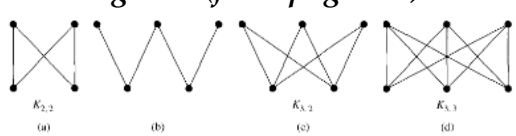


Figure 6 ▲

solution. Tree (a) in Figure 6 is isomorphic to T_1, Tree (b) is isomorphic to T_5, and Tree (c) is isomorphic to T_11. □

Problem 6. *Find two nonisomorphic spanning trees of $K_{3,3}$ drawn in Figure 7 on page 256. Explain why they are non-isomorphic.*

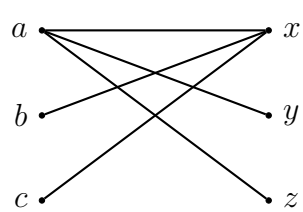
Figure 7 (from page 256):



Solution.

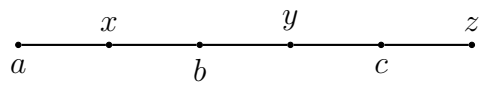
Two non-isomorphic spanning trees of $K_{3,3}$:

Tree 1: Double Star



Degree sequence: $(1, 1, 1, 1, 3, 3)$. Vertices a and x have degree 3; vertices b, c, y, z are leaves.

Tree 2: Path



Degree sequence: $(1, 1, 2, 2, 2, 2)$. Forms the path a-x-b-y-c-z.

The degree sequences are different. Tree 1 has degree sequence $(1, 1, 1, 1, 3, 3)$ with two vertices of degree 3, while Tree 2 has degree sequence $(1, 1, 2, 2, 2, 2)$ with no vertices of degree 3. Since isomorphic graphs must have identical degree sequences, these trees are non-isomorphic.

□

Problem 7. *Consider a tree with n vertices. It has exactly $n - 1$ edges [Lemma 2], so the sum of the degrees of its vertices is $2n - 2$ [Theorem 3 on page 230].*

(a) A certain tree has two vertices of degree 4, one vertex of degree 3, and one vertex of degree 2. If the other vertices have degree 1, how many vertices are there in the graph? Hint: If the tree has n vertices, $n - 4$ of them will have to have degree 1.

(b) Draw a tree as described in part (a).

Solution.

(a) Let n be the total number of vertices. We have 2 vertices of degree 4, 1 of degree 3, 1 of degree 2, and the remaining $n - 4$ vertices of degree 1.

The sum of degrees equals $2n - 2$:

$$2(4) + 1(3) + 1(2) + (n - 4)(1) = 2n - 2$$

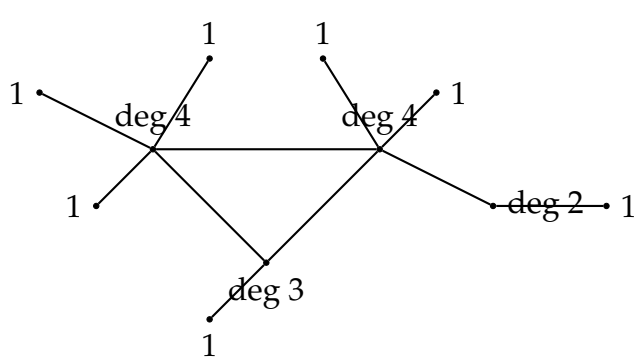
$$8 + 3 + 2 + n - 4 = 2n - 2$$

$$9 + n = 2n - 2$$

$$11 = n$$

The tree has 11 vertices.

(b)



□

Problem 8. *Repeat Exercise 7 for a tree with two vertices of degree 5, three of degree 3, two of degree 2, and the rest of degree 1.*

(a) How many vertices are there in the graph?

(b) Draw a tree as described.

Solution.

(a) Let n be the total number of vertices. We have 2 vertices of degree 5, 3 of degree 3, 2 of degree 2, and the remaining $n-7$ vertices of degree 1.

The sum of degrees equals $2n-2$:

$$2(5)+3(3)+2(2)+(n-7)(1)=2n-2$$

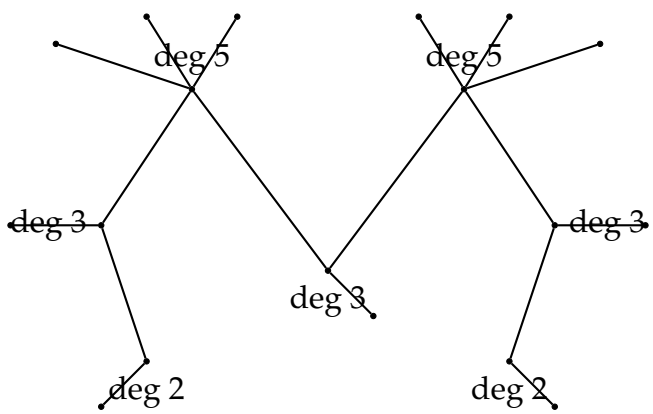
$$10+9+4+n-7=2n-2$$

$$16+n=2n-2$$

$$18=n$$

The tree has 18 vertices.

(b)

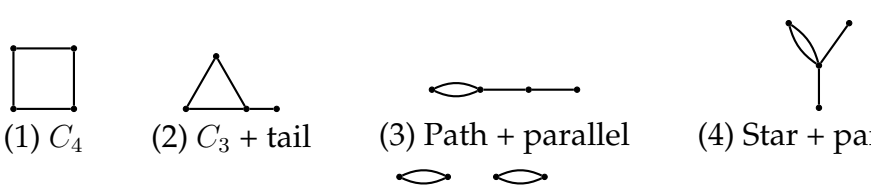


□

Problem 10. *Draw pictures of the nine connected graphs with four edges and four vertices. Don't forget loops and parallel edges.*

Solution.

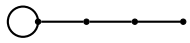
The nine connected graphs with 4 vertices and 4 edges are:



(1) C_4 (2) C_3 + tail (3) Path + parallel (4) Star + parallel

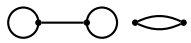
(5) Two pairs

(6) C_3 + loop



(7) Path + loop

(8) Star + loop



(9) Edge + loops + parallel

□

Problem 11. *(a) Show that a forest with n vertices and m components has $n-m$ edges.*

Solution.

(a) A forest is a disjoint union of trees. Let the m components be trees $T_1, T_2, \ldots, T_m$ with $n_1, n_2, \ldots, n_m$ vertices respectively.

Since each tree T_i has n_i vertices, it has exactly $n_i - 1$ edges (by Lemma 2).

The total number of vertices is:

$$n = n_1 + n_2 + \cdots + n_m$$

The total number of edges is:

$$e = (n_1 - 1) + (n_2 - 1) + \cdots + (n_m - 1)$$

$$e = (n_1 + n_2 + \cdots + n_m) - m$$

$$e = n - m$$

Therefore, a forest with n vertices and m components has $n - m$ edges.

□